

Sarang Shigwan

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Summary

AI-enabled Backend Engineer skilled in building **scalable APIs, data-driven systems, and production-ready services** using Python and modern web frameworks. Experienced in designing database schemas, handling concurrent workloads, and developing end-to-end systems with a focus on **performance, reliability, and clean architecture**.

Education

- **B.E. ENTC** at Marathwada Mitra Mandal's College of Engineering, Pune
Nov 2022 – May 2026 | **CGPA: 9.24 / 10**
- **H.S.C.** at Modern College of Arts, Science & Commerce, Pune
2020 – 2022 | **74.17%**
- **S.S.C.** at St. Francis De Sales School, Pune
2007 – 2020 | **89.80%**

Experience

Backend Intern(AWS Cloud) - EvolvingX Pvt. Ltd.

Oct 2024 – June 2025

- Developed and maintained a full-stack web platform using React (frontend) and Django (backend), enabling dynamic user interactions and RESTful API integration.
- Deployed the complete application on AWS EC2, with production environment configuration, HTTPS setup, and database integration.
- Optimized backend performance and ensured secure API communication through token validation and middleware customization.

Projects

LexVault (Smart Legal Document Manager)

Github Repo: [Lexvault](#)

- Built a full-stack document version control system ensuring immutable audit history and version traceability.
- Designed a PostgreSQL schema for version indexing and efficient retrieval of document histories.
- Developed REST APIs and a React dashboard to manage document lifecycles and timelines.
- Optimized backend performance for handling large document comparisons and concurrent updates

Skills : FastAPI, PostgreSQL, SQL, REST API Design, Database Schema Design, Version Control Systems, Data Modeling.

TraceLens

Github Repo: [Tracelens](#) | Design Report : [Report](#)

- Built a cloud-native system for automated root cause analysis of distributed microservice failures using a RAG pipeline with sentence-transformer embeddings and zero LLM dependency.
- Designed a stateless FastAPI service integrating Logs for cross-service trace correlation, deployable locally or on AWS Lambda.
- Implemented a vector similarity-based failure pattern engine with confidence scoring and actionable remediation.

Skills : Sentence-Transformers (all-MiniLM-L6-v2), RAG, NumPy, Pydantic, Vector Similarity Search, Cosine Similarity, NLP Embeddings

HFT Latency Monitoring Platform

Github Repo: [LatencyMonitor](#) | Design Report : [Report](#)

- Built a sub-millisecond UDP ingestion pipeline with P50/P95/P99 latency tracking on Kubernetes.
- Implemented Prometheus histograms and Grafana dashboards for real-time performance observability.
- Containerized a multi-service Go architecture using Docker and Kubernetes, debugging runtime issues for reliable deployments.

Skills : Low-latency systems, Go concurrency, Docker, Kubernetes, Prometheus/Grafana, Linux debugging, UDP pipelines.

Skills

- **Fundamentals :** Operating Systems, DBMS, Networking, OOP, DSA, SDLC, system design
- **Software Development :** Python(Django, FASTAPI), API Design, Microservices, React
- **Cloud/DevOps :** AWS (EC2, S3, VPC, Lambda, IAM), Terraform, Docker, GitHub Actions, CI/CD pipelines
- **GenAI & Intelligent Systems :** LLM integration, (RAG), vector embeddings, structured prompt engineering
- **Programming & Systems:** Python(Automation), Bash(Scripting), Go(basics), Linux, Git, SQL

Achievements

- **AIR 2** SAENIS Efficycle 2024
- **Winner** at Hackathons like Udaan 2023 and Dextirity 2024
- **Presented research papers at conferences like ICEI and GConSoft**